
Bayesian Inference

Application-Based Practice Set

A collection of challenging problems designed to develop intuition, derivation skills, and applied Bayesian reasoning.

Coverage

Topic 1	Classical vs Bayesian Inference
Topic 2	Bayes' Theorem and Prior Sensitivity
Topic 3	Likelihood Principle and Stopping Rules
Topic 4	Bayesian Data Analysis and Posterior Odds
Topic 5	Conditional Independence and Exchangeability
Topic 6	de Finetti's Representation
Topic 7	Beta–Bernoulli Model and Prediction
Topic 8	Conjugate Priors and Posterior Concentration
Topic 9	Normal–Normal Conjugacy
Topic 10	Marginal and Predictive Distributions
Topic 11	Improper Priors
Topic 12	Jeffreys Prior and Reparameterization
Topic 13	Kullback–Leibler Divergence and Fisher Information

Instructions: Attempt each question fully before checking any notes. Focus on the reasoning behind every step, not only the final answer.

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Chapter 1

Classical and Bayesian Foundations

1.1 Classical vs Bayesian Inference

Question 1.1 — The Meaning of Uncertainty

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1).$$

After observing the data, a researcher obtains

$$\bar{X} = 2.3, \quad n = 100.$$

(a) Construct the classical 95% confidence interval for θ .

(b) Explain precisely why the statement

$$P(\theta \in CI \mid X) = 0.95$$

is not a valid frequentist interpretation.

(c) Now suppose a Bayesian analysis uses

$$\theta \sim N(0, 100).$$

Derive the posterior distribution.

(d) Construct a 95% posterior credible interval.

(e) Explain carefully why the Bayesian credible interval permits a probability statement about θ after observing the data, whereas the classical confidence interval does not.

(f) As $n \rightarrow \infty$, compare the role played by the prior in the Bayesian analysis with the role played by repeated sampling in the frequentist interpretation.

Question 1.2 — Statistical vs Practical Significance

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1).$$

A scientist believes that effects satisfying

$$|\theta| < 0.01$$

are scientifically negligible.

Assume that

$$\theta = 0.002$$

and

$$n = 10^8.$$

- (a) Compute approximately the classical test statistic for testing

$$H_0 : \theta = 0 \quad \text{vs} \quad H_1 : \theta \neq 0.$$

- (b) Determine whether the classical test at level 0.05 would reject H_0 .

- (c) Explain why rejection of H_0 does not imply practical importance.

- (d) Suppose instead that the Bayesian quantity of interest is

$$P(|\theta| < 0.01 \mid X).$$

Explain why this question is more directly aligned with the scientific objective.

- (e) Give an example of a decision rule based on posterior probabilities that distinguishes practical significance from statistical significance.

Question 1.3 — Random Sample Size

Suppose the sample size N is determined by a preliminary coin toss:

$$N = \begin{cases} 2, & \text{with probability } 1/2, \\ 1000, & \text{with probability } 1/2. \end{cases}$$

Conditional on $N = n$,

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

The estimator is

$$\bar{X}_N.$$

- (a) Compute

$$E(\bar{X}_N \mid N).$$

- (b) Use the law of total variance to derive

$$\text{Var}(\bar{X}_N).$$

- (c) Explain why this unconditional variance may be misleading for describing the precision of the estimator after the experiment has actually been performed.

- (d) Suppose you observe that $N = 1000$. Which variance is relevant for inference about μ ?

- (e) Discuss the connection between this example and the distinction between pre-data and post-data uncertainty.

1.2 Bayes' Theorem and Prior Sensitivity

Question 2.1 — Screening Test with Prior Sensitivity

A disease has prevalence

$$P(D) = p.$$

A diagnostic test satisfies

$$P(T^+ | D) = 0.99, \quad P(T^+ | D^c) = 0.05.$$

- (a) Derive a general expression for

$$P(D | T^+)$$

as a function of p .

- (b) Evaluate this posterior probability when

$$p = 0.001, \quad p = 0.01, \quad p = 0.10, \quad p = 0.50.$$

- (c) Find the value of p for which

$$P(D | T^+) = \frac{1}{2}.$$

- (d) Explain why a highly accurate test can still have a low posterior probability of disease after a positive result.

- (e) Suppose the patient receives two conditionally independent positive test results. Derive

$$P(D | T_1^+, T_2^+).$$

- (f) Compare the effect of the second test with the effect of changing the prior prevalence.

Question 2.2 — Sequential Bayesian Updating

Suppose there are two hypotheses,

$$H_1, \quad H_2,$$

with prior probabilities

$$P(H_1) = 0.2, \quad P(H_2) = 0.8.$$

Three independent observations are collected. Their likelihood ratios are

$$\frac{p(y_1 | H_1)}{p(y_1 | H_2)} = 5,$$

$$\frac{p(y_2 | H_1)}{p(y_2 | H_2)} = \frac{1}{4},$$

and

$$\frac{p(y_3 | H_1)}{p(y_3 | H_2)} = 10.$$

- (a) Compute the prior odds.

- (b) Update the odds after observing y_1 .
- (c) Update again after observing y_2 .
- (d) Update again after observing y_3 .
- (e) Compute the final posterior probability of H_1 .
- (f) Show that updating sequentially gives the same answer as using the joint likelihood of all three observations.
- (g) Explain why the posterior probability does not necessarily move monotonically toward one hypothesis as observations arrive.

Question 2.3 — Three-Prisoners Generalization

Three prisoners A, B, C have exactly one pardon.
The prior probabilities are not necessarily equal:

$$P(A) = a, \quad P(B) = b, \quad P(C) = c,$$

where

$$a + b + c = 1.$$

Prisoner A asks the guard to name one prisoner among B and C who will be executed.
Suppose that:

- If A is pardoned, the guard names B with probability q and C with probability $1 - q$.
- If B is pardoned, the guard must name C .
- If C is pardoned, the guard must name B .

Let G_B denote the event that the guard names B .

- (a) Derive

$$P(A \mid G_B)$$

in terms of a, b, c, q .

- (b) Find the corresponding expression for

$$P(C \mid G_B).$$

- (c) Determine the conditions under which observing G_B provides no information about whether A is pardoned.
- (d) Specialize your answer to

$$a = b = c = \frac{1}{3}, \quad q = \frac{1}{2}.$$

- (e) Explain why treating the guard's statement as if it were a random event independent of the pardon assignment gives the wrong answer.

1.3 Likelihood Principle and Stopping Rules

Question 3.1 — Two Experiments

Consider Bernoulli observations with success probability p .

Experiment A fixes

$$n = 20$$

in advance and observes

$$x = 14$$

successes.

Experiment B continues sampling until the 14th success occurs and observes that the total number of trials required is 20.

(a) Write the likelihood function $L_A(p)$.

(b) Write the likelihood function $L_B(p)$.

(c) Show that

$$L_A(p) \propto L_B(p).$$

(d) Under a prior

$$p \sim \text{Beta}(\alpha, \beta),$$

derive the posterior distribution under both experiments.

(e) Show that the two posterior distributions are identical.

(f) Explain why a frequentist analysis may nevertheless produce different p -values.

(g) Discuss whether the stopping rule should influence a Bayesian inference about p in this situation.

Question 3.2 — Constructing Proportional Likelihoods

Suppose

$$X \sim \text{Poisson}(\lambda).$$

Consider two different experiments.

Experiment A: Observe $n = 5$ independent Poisson observations whose sum is

$$S = 12.$$

Experiment B: Observe Poisson events sequentially until the cumulative count reaches at least 12, and suppose the resulting observed likelihood is proportional to

$$\lambda^{12} e^{-5\lambda}.$$

(a) Derive the likelihood under Experiment A.

(b) Verify whether the likelihoods are proportional.

(c) Suppose

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

under a shape–rate parameterization. Derive the posterior under Experiment A.

- (d) Derive the posterior under Experiment B.
- (e) Explain which parts of the data-generating mechanism are relevant for Bayesian updating.
- (f) Give a careful interpretation of the statement:

“The likelihood contains all information about λ relevant to inference.”

Question 3.3 — A Stopping Rule Thought Experiment

A researcher observes Bernoulli trials and chooses one of the following stopping rules:

1. Stop after 100 observations.
2. Stop after observing 60 successes.
3. Stop when a likelihood ratio exceeds 100.

Suppose all three experiments eventually produce a data sequence containing exactly
60 successes and 40 failures.

- (a) Write the likelihood for p based only on the observed sequence.
- (b) Show that the likelihood has the form

$$L(p) \propto p^{60}(1-p)^{40}.$$

- (c) Under

$$p \sim \text{Beta}(2, 2),$$

derive the posterior.

- (d) Derive the posterior predictive probability of success on the next trial.
- (e) Explain why the three stopping rules do not change the posterior under the likelihood principle.
- (f) Give one practical situation where ignoring the stopping rule could nevertheless be problematic because the probabilistic model itself has been incorrectly specified.

Chapter 2

Bayesian Modeling and Dependence

2.1 The Three Steps of Bayesian Data Analysis

Question 4.1 — Building a Full Probability Model

A factory produces electronic components.

For each component,

$$Y_i = \begin{cases} 1, & \text{defective,} \\ 0, & \text{not defective.} \end{cases}$$

Assume

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

but the true defect probability θ is unknown.

Historical information suggests

$$\theta \sim \text{Beta}(2, 98).$$

A sample of 50 components contains 4 defective items.

- (a) Explicitly write the full joint probability model

$$p(\theta, y_1, \dots, y_{50}).$$

- (b) Identify the observable and unobservable quantities.
- (c) Derive the posterior distribution.
- (d) Compute the posterior mean.
- (e) Derive the posterior predictive probability that the next component is defective.
- (f) Describe a posterior predictive check that could be used to evaluate whether the Bernoulli model is plausible.
- (g) Explain why obtaining a posterior distribution is not the final step of Bayesian data analysis.

Question 4.2 — Model Misspecification

Suppose the previous factory data were collected over 10 different production days. Within each day, the defect probability may differ.

A proposed model is

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

However, an alternative model is

$$Y_{ij} \mid \theta_j \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta_j),$$

where j indexes the production day.

- (a) Explain the difference between the two probability models.
- (b) Under the first model, what type of dependence structure is induced marginally after integrating out θ ?
- (c) Why might posterior predictive checks reveal that the first model is inadequate?
- (d) Suggest a meaningful discrepancy statistic for comparing observed and replicated data.
- (e) Explain how Bayesian modeling separates the assumptions about the data-generating mechanism from the computational procedure used to obtain the posterior.

Question 4.3 — Prediction as Bayesian Inference

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2),$$

where σ^2 is known.

You place a prior

$$\theta \sim N(\mu_0, \tau_0^2).$$

- (a) Write the joint model for

$$\theta, Y_1, \dots, Y_n, \tilde{Y}.$$

- (b) Derive

$$p(\tilde{Y} \mid Y_1, \dots, Y_n)$$

using

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta, Y) p(\theta \mid Y) d\theta.$$

- (c) Explain why

$$p(\tilde{Y} \mid \theta, Y) = p(\tilde{Y} \mid \theta).$$

- (d) Compare the uncertainty in θ with the uncertainty in $\tilde{Y}$.

- (e) Explain why even if the posterior variance of θ approaches zero, the predictive distribution for $\tilde{Y}$ still retains observational variance.

2.2 Posterior Odds and Bayesian Evidence

Question 5.1 — Prior Odds vs Evidence

Consider

$$H_1 : \theta = \theta_1, \quad H_2 : \theta = \theta_2.$$

Suppose

$$\frac{P(H_1)}{P(H_2)} = \frac{1}{9}.$$

After observing data y ,

$$\frac{p(y | H_1)}{p(y | H_2)} = 50.$$

(a) Compute the posterior odds.

(b) Compute

$$P(H_1 | y).$$

(c) How large must the likelihood ratio be for

$$P(H_1 | y) > \frac{1}{2}?$$

(d) Explain how strong evidence from the data can overcome an initially unfavorable prior.

(e) Suppose instead that the prior odds are

$$\frac{1}{10^6}.$$

What likelihood ratio is required to obtain posterior odds greater than 1?

(f) Discuss the interpretation of extremely strong prior beliefs.

Question 5.2 — Posterior Odds Consistency

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} f(\cdot | \theta^*),$$

where

$$\theta^* \in \{\theta_1, \theta_2\}.$$

Define

$$R_n = \frac{p(Y_1, \dots, Y_n | \theta_1)}{p(Y_1, \dots, Y_n | \theta_2)}.$$

(a) Show that

$$\log R_n = \sum_{i=1}^n \log \frac{f(Y_i | \theta_1)}{f(Y_i | \theta_2)}.$$

- (b) Assume $\theta^* = \theta_1$. Apply the law of large numbers to

$$\frac{1}{n} \log R_n.$$

- (c) Show that the limiting expectation can be expressed using Kullback–Leibler divergence.
- (d) Under suitable identifiability conditions, explain why

$$R_n \rightarrow \infty$$

almost surely.

- (e) Deduce the asymptotic behavior of the posterior odds.
- (f) Repeat the argument when

$$\theta^* = \theta_2.$$

Question 5.3 — Conflicting Evidence

Suppose a model comparison begins with

$$P(H_1) = P(H_2) = \frac{1}{2}.$$

Three batches of data provide Bayes factors:

$$BF_{12}^{(1)} = 20, \quad BF_{12}^{(2)} = \frac{1}{5}, \quad BF_{12}^{(3)} = \frac{1}{2}.$$

- (a) Compute the posterior odds after all three batches.
- (b) Compute the posterior probability of H_1 .
- (c) Which batch contributes the strongest evidence?
- (d) Explain why a single strong piece of evidence can be offset by later evidence.
- (e) Show that

$$\log BF_{12}^{\text{total}} = \sum_{j=1}^3 \log BF_{12}^{(j)}.$$

- (f) Explain why the log-likelihood ratio is particularly useful for studying asymptotic Bayesian consistency.

2.3 Conditional Independence and Exchangeability

Question 6.1 — Conditionally Independent but Marginally Dependent

Suppose

$$\theta \sim \text{Beta}(\alpha, \beta),$$

and

$$Y_1, Y_2 \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

(a) Compute

$$P(Y_1 = 1).$$

(b) Compute

$$P(Y_1 = 1, Y_2 = 1).$$

(c) Determine

$$P(Y_2 = 1 \mid Y_1 = 1).$$

(d) Show that, in general,

$$P(Y_1 = 1, Y_2 = 1) \neq P(Y_1 = 1)P(Y_2 = 1).$$

(e) Compute

$$(Y_1, Y_2).$$

(f) Explain intuitively why observing Y_1 changes your prediction about Y_2 even though the observations are independent conditional on θ .

Question 6.2 — Exchangeable but Not iid

An urn contains exactly one red ball and one blue ball.

Two balls are drawn without replacement.

Let

$$Y_i = \begin{cases} 1, & \text{red,} \\ 0, & \text{blue.} \end{cases}$$

(a) Find the joint distribution of

$$(Y_1, Y_2).$$

(b) Show that

$$(Y_1, Y_2)$$

is exchangeable.

(c) Compute

$$P(Y_2 = 1).$$

(d) Compute

$$P(Y_2 = 1 \mid Y_1 = 1).$$

(e) Show that the observations are not independent.

(f) Explain why this example shows

$$\text{iid} \Rightarrow \text{exchangeable}$$

but not necessarily the converse.

(g) Discuss why finite exchangeability does not automatically imply the existence of an infinite iid-given- θ representation.

Question 6.3 — Predicting the Next Observation

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta).$$

After observing

$$z = \sum_{i=1}^n Y_i$$

successes:

(a) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(b) Derive

$$P(Y_{n+1} = 1 \mid \theta, Y_1, \dots, Y_n).$$

(c) Explain the difference between these two probabilities.

(d) Show explicitly that the past observations affect the predictive distribution only through the posterior distribution of θ .

(e) Under what conditions does

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n)$$

approach

$$\frac{z}{n}?$$

(f) Interpret this result as a form of Bayesian learning.

2.4 de Finetti's Representation

Question 7.1 — Beta Mixing Produces Exchangeability

Suppose

$$\theta \sim \text{Beta}(\alpha, \beta),$$

and conditional on θ ,

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

(a) Derive

$$P(Y_1 = y_1, \dots, Y_n = y_n)$$

by integrating out θ .

- (b) Show that the resulting expression depends on the data only through

$$z = \sum_{i=1}^n y_i.$$

- (c) Deduce that the sequence is exchangeable.
 (d) Show that the observations are not generally independent marginally.
 (e) Derive the marginal distribution of

$$Z = \sum_{i=1}^n Y_i.$$

- (f) Identify this distribution.

Question 7.2 — Infinite vs Finite Exchangeability

Consider a finite binary sequence

$$(Y_1, Y_2)$$

defined by

$$P(Y_1 = 1, Y_2 = 0) = \frac{1}{2},$$

$$P(Y_1 = 0, Y_2 = 1) = \frac{1}{2},$$

with all other joint probabilities equal to zero.

- (a) Show that (Y_1, Y_2) is exchangeable.
 (b) Show that the variables cannot be represented as conditionally iid Bernoulli observations under any mixing distribution on θ .
 (c) Hint: compute

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2).$$

- (d) Explain why this provides a warning against applying de Finetti's theorem to an arbitrary finite exchangeable collection.
 (e) State the role played by infinite extendability in the theorem.

Question 7.3 — Recovering a Mixing Distribution

Suppose an exchangeable binary sequence satisfies

$$P(Y_1 = 1) = \frac{1}{2},$$

and

$$P(Y_1 = 1, Y_2 = 1) = \frac{1}{3}.$$

Assume that the sequence has a de Finetti representation with

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

(a) Show that

$$E(\theta) = \frac{1}{2}.$$

(b) Show that

$$E(\theta^2) = \frac{1}{3}.$$

(c) Compute

$$\text{Var}(\theta).$$

(d) Compute

$$(Y_1, Y_2).$$

(e) Suppose $\theta \sim \text{Beta}(\alpha, \alpha)$. Determine α so that the above two moment conditions hold.

(f) Explain how dependence among observations provides information about uncertainty in the latent parameter θ .

Chapter 3

Conjugate Bayesian Models

3.1 Beta–Bernoulli Model

Question 8.1 — Prior Information as Pseudo-Observations

Suppose

$$Y_1, \dots, Y_{20} \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

and the observed number of successes is

$$z = 12.$$

Consider the following priors:

$$\text{Prior A: } \theta \sim \text{Beta}(1, 1),$$

$$\text{Prior B: } \theta \sim \text{Beta}(20, 20),$$

$$\text{Prior C: } \theta \sim \text{Beta}(100, 20).$$

- (a) Derive the posterior distribution under each prior.
- (b) Compute the posterior mean under each prior.
- (c) Compute the posterior predictive probability of success.
- (d) Compare the posterior variances qualitatively.
- (e) Interpret the prior parameters α and β as prior information about successes and failures.
- (f) Which prior has the strongest influence on the posterior?
- (g) Explain whether it is correct to call α and β literal historical observations.

Question 8.2 — Laplace's Rule of Succession

Suppose

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

with

$$\theta \sim \text{Beta}(1, 1).$$

A sequence of n observations contains z successes.

(a) Derive the posterior distribution of θ .

(b) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(c) Compare this quantity with the MLE

$$\hat{\theta}_{\text{MLE}} = \frac{z}{n}.$$

(d) Show algebraically that

$$\frac{z+1}{n+2} = \frac{n}{n+2} \frac{z}{n} + \frac{2}{n+2} \frac{1}{2}.$$

(e) Interpret the posterior mean as shrinkage toward $\frac{1}{2}$.

(f) Examine the cases

$$(z, n) = (0, 1), (1, 1), (1, 2), (50, 100).$$

(g) Explain why the difference between Bayesian prediction and the MLE becomes negligible for large samples.

Question 8.3 — Prior Predictive Distribution

Suppose

$$\theta \sim U(0, 1),$$

and conditional on θ ,

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

Let

$$Z = \sum_{i=1}^n Y_i.$$

(a) Derive

$$P(Z = z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta.$$

(b) Evaluate the integral using the Beta function.

(c) Show that

$$P(Z = z) = \frac{1}{n+1}, \quad z = 0, \dots, n.$$

(d) Explain why this result may initially seem surprising.

(e) Compare the prior predictive distribution under

$$\theta \sim U(0, 1)$$

with the sampling distribution under a fixed

$$\theta = \frac{1}{2}.$$

- (f) Explain the conceptual difference between uncertainty about θ and randomness conditional on θ .

3.2 Conjugacy and Posterior Concentration

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

and

$$\theta \sim \text{Beta}(\alpha, \beta).$$

Let

$$Z = \sum_{i=1}^n Y_i.$$

- (a) Derive the posterior distribution.
- (b) Derive the posterior mean.
- (c) Show that the posterior mean can be written as a weighted average of the prior mean and the sample proportion.
- (d) Derive the posterior variance.
- (e) Assume

$$\frac{Z}{n} \rightarrow \theta_0.$$

Show heuristically that the posterior variance approaches zero as $n \rightarrow \infty$.

- (f) Explain why posterior concentration does not necessarily mean that every prior leads to the same posterior for finite n .

Question 9.3 — Exponential Family and Conjugacy

The Binomial likelihood is

$$P(X = x \mid p) = \binom{n}{x} p^x (1-p)^{n-x}.$$

- (a) Rewrite the likelihood in exponential-family form.
- (b) Identify a natural parameter.
- (c) Identify a sufficient statistic.
- (d) Starting from a general exponential-family conjugate prior form, show why a Beta prior is conjugate.

- (e) Derive the posterior hyperparameter update.
- (f) Explain why conjugacy is computationally convenient.
- (g) Discuss one limitation of restricting Bayesian modeling only to conjugate families.

3.3 Normal–Normal Conjugacy

Question 10.1 — Precision-Weighted Updating

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2),$$

where

$$\sigma^2 = 4.$$

The prior is

$$\theta \sim N(10, 9).$$

For

$$n = 25, \quad \bar{Y} = 12,$$

answer the following.

- (a) Derive the posterior variance.
- (b) Derive the posterior mean.
- (c) Express the posterior mean as

$$w\mu_0 + (1 - w)\bar{Y}.$$

- (d) Compute w .
- (e) Explain why the observation summary $\bar{Y}$ receives more weight than the prior mean.
- (f) Determine what happens to the posterior mean and variance as

$$n \rightarrow \infty.$$

- (g) Determine what happens if

$$\tau_0^2 \rightarrow 0.$$

Question 10.2 — Sequential Normal Updating

Suppose

$$\theta \sim N(\mu_0, \tau_0^2),$$

and observations arrive one at a time:

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2).$$

- (a) Derive the posterior after observing Y_1 .

- (b) Treat this posterior as the prior for observing Y_2 and derive the next posterior.
- (c) Continue conceptually until n observations have been observed.
- (d) Show that the resulting posterior precision is

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

- (e) Show that sequential updating gives the same posterior as analyzing all observations simultaneously.
- (f) Explain why Bayesian updating can be performed online without storing all previous raw observations when sufficient statistics are available.

Question 10.3 — Prior-Data Conflict

Suppose

$$\theta \sim N(0, 1),$$

and

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, 1).$$

You observe

$$n = 5, \quad \bar{Y} = 10.$$

- (a) Derive the posterior distribution.
- (b) Compute the posterior mean.
- (c) Explain why the posterior mean is neither 0 nor 10.
- (d) Now suppose the prior variance changes from 1 to 100. Derive the new posterior.
- (e) Compare the two posterior means.
- (f) Explain how increasing prior variance represents weaker prior certainty.
- (g) Discuss whether a prior that is strongly contradicted by the data should automatically be discarded.

3.4 Marginal and Predictive Distributions

Question 11.1 — Marginal Distribution

Suppose

$$\theta \sim N(\mu_0, \tau_0^2),$$

and

$$Y \mid \theta \sim N(\theta, \sigma^2).$$

- (a) Write

$$Y = \theta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \sigma^2).$$

(b) Derive the marginal distribution of Y .

(c) Compute

$$E(Y).$$

(d) Compute

$$\text{Var}(Y)$$

using the law of total variance.

(e) Show that

$$\text{Var}(Y) = \tau_0^2 + \sigma^2.$$

(f) Interpret the two components of the marginal variance.

Question 11.2 — Dependence in a Hierarchical Normal Model

Suppose

$$Y_i = \theta + \varepsilon_i,$$

where

$$\theta \sim N(\mu_0, \tau_0^2),$$

and

$$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

(a) Compute

$$E(Y_i).$$

(b) Compute

$$\text{Var}(Y_i).$$

(c) Compute

$$(Y_i, Y_j), \quad i \neq j.$$

(d) Compute the correlation between two observations.

(e) Explain why the observations are independent conditional on θ but correlated marginally.

(f) Discuss how this normal example is analogous to the Beta–Bernoulli model.

Question 11.3 — Posterior Predictive Distribution

Suppose after observing data Y you obtain

$$\theta \mid Y \sim N(\mu_n, \tau_n^2).$$

A future observation satisfies

$$\tilde{Y} \mid \theta \sim N(\theta, \sigma^2).$$

(a) Derive

$$E(\tilde{Y} \mid Y).$$

(b) Derive

$$\text{Var}(\tilde{Y} \mid Y).$$

(c) Hence identify the posterior predictive distribution.

(d) Explain why

$$\text{Var}(\tilde{Y} \mid Y) > \text{Var}(\theta \mid Y).$$

(e) Suppose

$$\tau_n^2 \rightarrow 0.$$

What does the predictive distribution converge to?

(f) Explain the difference between estimating θ accurately and predicting an individual future observation accurately.

Chapter 4

Objective Bayesian Analysis

4.1 Improper Priors

Question 12.1 — When an Improper Prior Works

Suppose

$$X_1, \dots, X_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2),$$

where σ^2 is known.

Use the improper prior

$$\pi(\theta) \propto 1, \quad -\infty < \theta < \infty.$$

(a) Show that the prior cannot be normalized.

(b) Derive the posterior kernel.

(c) Show that the posterior is proper.

(d) Identify the posterior distribution.

(e) Explain why

$$\pi(\theta) \propto 1$$

is sometimes interpreted as a limiting case of

$$\theta \sim N(0, \tau^2)$$

as

$$\tau^2 \rightarrow \infty.$$

(f) Explain why the propriety of the posterior must be checked rather than assumed.

Question 12.2 — When an Improper Prior Fails

Suppose

$$X \mid \theta \sim U(0, \theta), \quad \theta > 0.$$

Consider the improper flat prior

$$\pi(\theta) \propto 1.$$

(a) Write the likelihood.

(b) Derive the posterior kernel.

(c) Determine whether

$$\int_0^\infty \pi(\theta)p(x | \theta) d\theta$$

is finite.

(d) Conclude whether the posterior is proper.

(e) Now consider

$$\pi(\theta) \propto \frac{1}{\theta}.$$

(f) Derive the corresponding posterior.

(g) Determine whether it is proper.

(h) Explain why “noninformative” does not mean that every improper prior is automatically acceptable.

Question 12.3 — Diffuse Proper vs Improper

Suppose

$$Y | \theta \sim N(\theta, 1).$$

Compare the priors

$$\theta \sim N(0, 10),$$

$$\theta \sim N(0, 10^6),$$

and

$$\pi(\theta) \propto 1.$$

(a) Derive the posterior distribution under each proper prior.

(b) Determine the limiting posterior as the prior variance tends to infinity.

(c) Compare the limiting posterior with the posterior obtained using the improper flat prior.

(d) Explain why a highly diffuse normal prior is not exactly the same mathematical object as an improper prior.

(e) Discuss whether increasing the variance of a proper prior always makes it “less informative” under every possible reparameterization.

4.2 Jeffreys Prior and Reparameterization

Question 13.1 — Bernoulli Jeffreys Prior

Suppose

$$Y \mid p \sim \text{Bernoulli}(p).$$

- (a) Write the log-likelihood.
- (b) Compute the first derivative with respect to p .
- (c) Compute the second derivative.
- (d) Derive the Fisher information

$$I(p) = -\mathbb{E} \left[\frac{\partial^2}{\partial p^2} \log f(Y \mid p) \right].$$

- (e) Derive Jeffreys prior

$$\pi_J(p) \propto \sqrt{I(p)}.$$

- (f) Identify the corresponding Beta distribution.
- (g) Compare this prior with the uniform prior

$$\text{Beta}(1, 1).$$

- (h) Explain why the two priors represent different notions of “noninformativeness.”

Question 13.2 — Flat Priors Are Not Invariant

Suppose

$$p \in (0, 1)$$

and define the log-odds parameter

$$\phi = \log \frac{p}{1-p}.$$

- (a) Derive

$$p = \frac{e^\phi}{1 + e^\phi}.$$

- (b) Compute

$$\frac{dp}{d\phi}.$$

- (c) Suppose

$$\pi(p) \propto 1.$$

Derive the induced prior density for ϕ .

- (d) Suppose instead

$$\pi(\phi) \propto 1.$$

Derive the induced prior density for p .

- (e) Show explicitly that flatness is not preserved under transformation.
- (f) Explain how Jeffreys prior addresses this issue.

Question 13.3 — Jeffreys Prior for a Normal Mean

Suppose

$$Y \mid \mu \sim N(\mu, \sigma^2),$$

where σ^2 is known.

- (a) Derive the Fisher information for μ .
- (b) Obtain Jeffreys prior.
- (c) Determine whether the prior is proper.
- (d) Compare it with the commonly used flat prior

$$\pi(\mu) \propto 1.$$

- (e) Explain why they coincide in this location model.
- (f) Now consider a transformed parameter

$$\phi = a\mu + b, \quad a \neq 0.$$

Show how Jeffreys prior transforms.

- (g) Discuss why location models behave differently from bounded-parameter models such as the Bernoulli model.

4.3 Kullback–Leibler Divergence and Fisher Information

Question 14.1 — KL Divergence for Bernoulli Models

Let

$$P = \text{Bernoulli}(p), \quad Q = \text{Bernoulli}(q).$$

- (a) Derive

$$D_{\text{KL}}(P \parallel Q).$$

- (b) Show that

$$D_{\text{KL}}(P \parallel Q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

- (c) Verify that

$$D_{\text{KL}}(P \parallel P) = 0.$$

- (d) Explain why

$$D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$$

in general.

(e) Let

$$q = p + \delta.$$

Use a Taylor expansion to obtain the second-order approximation of the KL divergence for small δ .

(f) Show how the Fisher information

$$I(p) = \frac{1}{p(1-p)}$$

appears in the approximation.

Question 14.2 — Local Distinguishability

Suppose

$$Y \mid \theta \sim f(y \mid \theta).$$

Define

$$D(\theta, \theta + d\theta) = D_{\text{KL}}(f(\cdot \mid \theta) \parallel f(\cdot \mid \theta + d\theta)).$$

(a) Write the log-density expansion

$$\log f(y \mid \theta + d\theta)$$

up to second order.

(b) Take expectation under

$$Y \sim f(\cdot \mid \theta).$$

(c) Show why the first-order term vanishes.

(d) Use the definition of Fisher information to obtain a quadratic approximation involving

$$I(\theta)(d\theta)^2.$$

(e) Explain why Fisher information can be interpreted as a local measure of distinguishability between nearby parameter values.

(f) Explain how this interpretation motivates

$$\pi_J(\theta) \propto \sqrt{I(\theta)}.$$

Question 14.3 — KL Divergence and Posterior Consistency

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} f(\cdot \mid \theta_0).$$

Consider two parameter values

$$\theta_0 \quad \text{and} \quad \theta_1.$$

Define the log-likelihood ratio

$$L_n = \log \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_0)}.$$

(a) Show that

$$L_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_0)}.$$

(b) Divide by n and apply the law of large numbers.

(c) Show that the limiting expectation can be written as

$$-D_{\text{KL}}(f(\cdot \mid \theta_0) \parallel f(\cdot \mid \theta_1)).$$

(d) Assuming the KL divergence is strictly positive whenever

$$\theta_1 \neq \theta_0,$$

show that

$$L_n \rightarrow -\infty$$

at an approximately linear rate.

(e) Deduce that the likelihood ratio in favor of θ_1 tends to zero.

(f) Explain how this argument provides intuition for posterior concentration around the true parameter.

(g) Discuss the assumptions required before claiming full Bayesian consistency.

Chapter 5

Integrated Challenge Problems

5.1 Full Bayesian Analysis Problems

Integrated Challenge 1 — A Complete Beta–Bernoulli Analysis

A new online platform believes that the probability θ that a user clicks an advertisement is unknown.

Historical data suggest

$$\theta \sim \text{Beta}(4, 16).$$

A new experiment observes

$$n = 50$$

users and

$$z = 15$$

clicks.

- (a) Write the full probability model.
- (b) Derive the posterior distribution.
- (c) Compute the posterior mean and posterior mode.
- (d) Derive the posterior variance.
- (e) Compare the posterior mean with the MLE.
- (f) Derive the posterior predictive probability of a click from the next user.
- (g) Derive the predictive distribution for the number of clicks among the next 10 users.
- (h) Explain how you would perform a posterior predictive check.
- (i) Compare the result with an analysis using the uniform prior.
- (j) Discuss whether the prior information remains important after observing 50 users.

Integrated Challenge 2 — Competing Normal Models

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\theta, 1).$$

You are comparing

$$H_0 : \theta = 0$$

against

$$H_1 : \theta = 1.$$

The prior probabilities are

$$P(H_0) = 0.9, \quad P(H_1) = 0.1.$$

Suppose

$$n = 20, \quad \bar{Y} = 0.7.$$

(a) Derive the likelihood ratio

$$\frac{p(Y | H_1)}{p(Y | H_0)}.$$

(b) Compute the posterior odds.

(c) Compute the posterior probabilities.

(d) Determine whether the data overcome the strong prior preference for H_0 .

(e) Derive the expression for the log-likelihood ratio for general n and $\bar{Y}$.

(f) Assume the true parameter is $\theta = 1$. Use the law of large numbers to explain what happens to the posterior odds as $n \rightarrow \infty$.

(g) Interpret the limiting behavior using KL divergence.

Integrated Challenge 3 — From Exchangeability to Prediction

Suppose

$$Y_1, Y_2, \dots$$

is assumed to be an infinitely exchangeable binary sequence.

By de Finetti's theorem, assume that there exists a latent random variable θ such that

$$Y_i | \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

Further suppose

$$\theta \sim \text{Beta}(\alpha, \beta).$$

After observing

$$n$$

observations with

$$z$$

successes:

(a) Derive the joint marginal distribution of the observed sequence.

(b) Show that the sequence is exchangeable.

(c) Derive the posterior distribution of θ .

(d) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(e) Derive

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid Y_1, \dots, Y_n).$$

(f) Show that the future observations remain dependent after integrating out θ .

(g) Compute the posterior predictive covariance between

$$Y_{n+1}$$

and

$$Y_{n+2}.$$

(h) Explain how this single model simultaneously illustrates:

- exchangeability,
- conditional independence,
- marginal dependence,
- Bayesian updating,
- posterior prediction,
- conjugacy.

How to Use This Practice Set

For every question, try to follow this workflow:

Model $\rightarrow$ Prior $\rightarrow$ Likelihood $\rightarrow$ Posterior $\rightarrow$ Inference / Prediction

Do not immediately search for a formula.

Instead ask:

1. What are the unknown quantities?
2. What is random?
3. What assumptions define the probability model?
4. What information is contained in the likelihood?
5. What information is represented by the prior?
6. How does Bayes' rule combine them?
7. What quantity do I actually want after obtaining the posterior?

Recommended Difficulty Strategy

First pass: Solve without notes.

Second pass: If stuck, identify exactly which concept is missing.

Third pass: Redo the derivation from first principles.

Final goal: You should eventually be able to look at a new problem and independently decide:

What is the model? What is the prior? What is the likelihood? What does the posterior answer?